Name: Student ID:

There is a data set built into R called **`mtcars`** that includes several measures on different types of cars. Learn more about the data set using the **`?mtcars`** command (many datasets have help pages!).

Overall goal: We want to estimate the *fuel efficiency* of cars based on their *weight*. We're going to explore this in several steps.

Question 1

part a

One way to "summarize" a relationship is simply to plot it. "Summarize" the association between the fuel efficiency and the weight of the car by making a scatter plot.

- **Remember**: Run the command **`?mtcars`** to learn what all the columns mean.
- **Remember**: The variable we are *modeling* (estimating) always goes on the y axis.

part b

Calculate the value of the correlation coefficient between fuel efficiency and weight below using `dplyr` code. Copy the code below, as well as the value of the correlation coefficient.

Question 2

part a

Fit a linear model using `lm` to estimate fuel efficiency using weight. Write the code you used below. Store the linear model in an object called `q2`.

part b

Based on the model output, write down the mathematical equation of the linear model.

Question 3

part a

Repeat **Question 1, part a** but use the horsepower of the car instead of the weight as the explanatory variable. You do not need to copy the code below.

part b

Calculate the value of the correlation coefficient between fuel efficiency and horsepower below using `dplyr` code. Copy the code below, as well as the value of the correlation coefficient.

part c

Repeat **Question 2 (parts a and b)**. You do not need to copy your code, but do write down the mathematical form of the linear model used to explain fuel efficiency with horsepower.

Question 4

It seems both weight and horsepower are related to fuel efficiency. How do we determine which relationship is stronger? Do we look at A) the correlation coefficients or B) the slopes of the linear models? Why? **Explain in 1-2 sentences.**

Question 5

We expect heavier cars to have worse fuel efficiency (it takes more gas to move a bigger car, of course). But which car has the *worst* fuel efficiency *given its weight*? Think for a moment about what that question means, then write code to find the answer. **State the car and provide supporting code for your answer.**